

# ARIA: AN ANALYTIC SOURCE-FILTER NEURAL VOCODER FOR INTERPRETABLE, DECOUPLED PHONETIC CONTROL

TODO Author(s)

TODO Affiliation

## ABSTRACT

We present ARIA, an analytic, differentiable source-filter neural vocoder for interpretable and decoupled control of the acoustic parameters phoneticians reason with. Unlike black-box neural vocoders, ARIA’s network only estimates per-frame interpretable parameters— $F_0$ , formants  $F_1/F_2$  with bandwidths, spectral tilt, energy, and a glottal voice-quality parameter ( $R_d$ )—while a fully deterministic DSP model realises them via closed-form analytic pole placement and a Liljencrants–Fant glottal wavetable, pushing the glass-/black-box boundary to the encoder and giving control that is decoupled by construction. On single-speaker,  $\approx 1$  h/voice models in Mandarin and English, ARIA attains near-exact manipulation (median error  $\approx 0.9$  Hz for  $F_0$ ,  $\approx 20$  Hz for  $F_2$ ) with clean  $F_0/F_1/F_2$  independence, and is released as an interactive web instrument: move one phonetic knob and only that knob moves.

Index Terms— controllable speech synthesis, neural vocoder, source-filter model, differentiable DSP, interpretability

## 1. INTRODUCTION

Research in phonetics and voice quality, and increasingly controllable text-to-speech, needs a synthesiser one can probe: move a single interpretable acoustic parameter—a fundamental frequency, an individual formant, a glottal voice-quality setting—and have only that property change, while reading its realised value back from the signal. Modern black-box neural vocoders synthesise with excellent quality but expose no such handles; classic source-filter and formant synthesisers expose exactly these handles, but sound dated.

We position ARIA on the black-box→glass-box spectrum that runs from DDSP [1], through neural source-filter models [2], to HiFi-Glot [3]. ARIA sits at the far glass-box end: the network only estimates a small set of interpretable parameters, and a fully deterministic, analytic differentiable-DSP source-filter model realises them.

Interactive demo: [TODO-demo-url](#). Audio in the demo is model output.

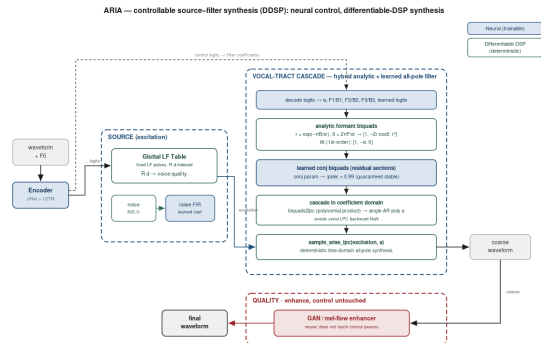


Fig. 1. ARIA pipeline: the encoder estimates interpretable per-frame parameters; a deterministic analytic source-filter decoder realises them.

Any adversarial training is a training-time gradient signal only and is absent from the inference path, so what is commanded is what is synthesised. ARIA is built on the differentiable-DSP machinery of GOLF [4], trading a measure of raw modelling capacity for exact, decoupled, physically meaningful control.

Our contributions are: (i) an analytic source-filter neural vocoder whose formant control is closed-form, range-bounded, and decoupled by construction; (ii) near-exact, directly measurable manipulation fidelity for  $F_0$  and  $F_2$  with clean  $F_0/F_1/F_2$  independence; (iii) a low-resource recipe—per-voice single-speaker models from  $\sim 1$  h of audio in Mandarin and English; and (iv) an open, interactive web instrument for phonetic exploration. We do not claim independent control of voice quality or energy (they co-vary), nor nasalisation, as the analytic tract is all-pole.

## 2. SYSTEM DESCRIPTION

### 2.1. Pipeline overview

ARIA is a source-filter neural vocoder in which a network estimates only interpretable acoustic parameters and a deterministic, parameter-free differentiable-DSP decoder turns them back into a waveform (Fig. 1). Following GOLF [4], the encoder is a U-Net ( $n_{\text{fft}}=512$ , hop

160) followed by a 3-layer BiLSTM ( $\approx 5.5$  M parameters). It consumes the analysis frames and the ground-truth  $F_0$  (pitch and voicing are taken from the analysis, not predicted) and emits, at a 10 ms hop, a control vector whose every channel is a named, range-constrained physical quantity — log gain, spectral tilt, formant centre frequencies and bandwidths, a glottal voice-quality index, and a few free residual coefficients — rather than an opaque latent.

The decoder chains a glottal-flow source, a band-wise harmonic/noise mixer, the analytic all-pole vocal tract, and a global room filter, with no learnable weights on the signal path: the waveform is a closed-form, differentiable function of physically meaningful quantities. The same vector is therefore both the training target and the inference-time control surface (Sec. 2.4); editing a track and re-running the decoder yields an exact, predictable change, unlike DDSP-style decoders [1, 2] whose envelopes are emitted directly and are not individually addressable.

## 2.2. Analytic vocal-tract control

The vocal tract is the central contribution. Rather than predicting a filter response and hoping formants emerge, ARIA places each resonance by a closed-form map from its centre frequency  $F$  and bandwidth  $B$  (Hz) to a conjugate pole pair. For sample rate  $f_s$ ,

$$r = \exp(-\pi B/f_s), \quad \theta = 2\pi F/f_s, \quad (1)$$

giving the second-order all-pole section

$$\frac{1}{A_i(z)} = \frac{1}{1 - 2r \cos \theta z^{-1} + r^2 z^{-2}}. \quad (2)$$

$F_1, F_2$  are produced this way under explicit Hz ranges ( $F_1 \in [150, 1300]$ ,  $F_2 \in [600, 3200]$ ) via a sigmoid map of the logits; spectral tilt is a single first-order real pole  $[1, -\alpha, 0]$  capturing the global rolloff without consuming a resonance. We append  $n_{\text{learned}}=6$  stability-bounded “free” pole pairs for higher resonances not exposed as knobs.

Because Eq. (1) is an analytic, per-formant map with no learned parameter  $\rightarrow$  pole stage,  $F_1$  and  $F_2$  are decoupled by construction: moving  $F_2$  moves exactly the second pole pair. For efficiency the sections are not cascaded as serial LPC filters (which explode in the backward pass); each biquad is composed in the coefficient domain into one AR polynomial  $A(z) = \prod_i A_i(z)$  via `biquads2lpc` and applied by a single time-varying `sample_wise_lpc` call (order 18 at 16 kHz).

## 2.3. Source and aperiodicity

The periodic excitation is a Liljencrants–Fant (LF) glottal-flow derivative wavetable [5]: a 2048-point ta-

ble over a log grid of the LF voice-quality parameter  $R_d \in [0.3, 2.7]$ , indexed by a single per-frame value, so glottal character (breathy  $\leftrightarrow$  pressed) is one interpretable scalar. The lookup is oversampled and windowed-sinc decimated to suppress the aliasing from frame-to-frame  $R_d$  change. A purely additive source can only raise the noise floor, leaving a high-band harmonic artifact; ARIA instead uses a decoupled band-aperiodicity head,

$$\text{src} = (\mathbf{1} - A) \text{harm} + A H(\text{noise}), \quad (3)$$

where  $H$  colours the aspiration noise and  $A(f) \in [0, 1]$  is an explicit band aperiodicity (inverse-HNR, WORLD/D4C sense [6]) crossfading the branches per band — letting high harmonics dissolve into aspiration while keeping aperiodicity itself editable. The source is then filtered by  $1/A(z)$  and a fixed room filter.

## 2.4. Control surface and manipulation API

Because every decoder input is a named physical quantity, the control vector fit during training is the editing interface, and it drives the demo directly. `get_formant_params` decodes logits to Hz; `set_formant_params` writes absolute targets back through the exact inverse range map (so “ $F_2=2200$  Hz” is reproduced to the Hz); `scale_formants` applies a relative, contour-preserving per-frame multiplier (e.g. “raise  $F_1$  by 15%”), and likewise covers tilt and energy. Each re-runs the deterministic decoder of Eqs. (1)–(3), so a manipulation is a clean re-synthesis from edited parameters with no auxiliary network in the loop.

## 3. THE DEMONSTRATION

The control surface of Sec. 2.4 is exposed as an open, dependency-light web instrument (static HTML/JavaScript, deployable to GitHub Pages; all audio is real ARIA synthesis), driven by a manifest of pre-rendered analysis-by-synthesis clips so a visitor explores the model with no server or GPU. In the manipulation studio one picks an utterance and a phonetic axis— $F_0$ ,  $F_1$  (openness),  $F_2$  (backness), glottal voice quality ( $R_d$ , modal  $\leftrightarrow$  breathy), or energy—and drags a slider through a seven-step continuum; the spectrogram, audio and read-back values update together, while a companion panel plots the  $F_1$ – $F_2$  trajectory (or  $F_0$  track) so the held parameters are visibly stationary as the swept one moves: moving one knob moves only that knob. Each step is a fresh re-synthesis from edited parameters (Sec. 2.4), closing the command  $\rightarrow$  realisation loop the visitor hears. The studio runs on F024 (Mandarin, 16 kHz, v4); a cross-lingual section repeats the  $F_0$ , joint  $F_1+F_2$  and voice-quality continua on CSMSC (Mandarin) and LJSpeech (English) from  $\sim 1$  h single-speaker models, showing the analytic

control surface transfers across speakers and languages at low resource.

## 4. EVALUATION

### 4.1. Parameter independence

ARIA’s controls are interpretable only if commanding one parameter does not silently move the others. We quantify this with a  $5 \times 5$  leakage matrix: each control is perturbed by  $\pm 10\%$  and the change in every acoustic parameter is measured, normalised so the on-diagonal response is 1. F0, F1 and F2 are mutually near-independent: on connected-speech CSMSC the cross-terms are negligible (F0→F1 0.01, F0→F2 0.00, F1→F2 0.03, F2→F1 0.00), and with supervised poles on F024 they stay small ( $\leq 0.24$ ).<sup>1</sup> We scope the independence claim to {F0, F1, F2}: spectral tilt and energy co-vary strongly with each other and the formants (e.g. CSMSC F1→tilt 2.70, F2→energy 1.36) and are excluded.

### 4.2. Manipulation fidelity

A commanded value must also be realised. The median absolute error  $|\text{measured} - \text{target}|$  over voiced frames is 0.9 Hz (F0), 14.6 Hz (F1) and 20.6 Hz (F2)—a small fraction of the natural spread ( $\sigma=31/84/290$  Hz), so F0 and F2 control are near-exact. F1 is the weaker knob: on F024 it under-shoots ( $R^2 \approx 0.78$ ,  $\approx 0.91$  with pole supervision, slope  $< 1$ ) but is strong on connected speech ( $R^2 \approx 0.96$ – $0.99$  on CSMSC/LJ). The  $\sim 20$  Hz F2 error compares favourably to the  $\sim 150$  Hz reported for HiFi-Glot [3].

### 4.3. Naturalness

Table 1 reports UTMOS [7]—a neural MOS proxy, not a human listening test—as mean  $\pm$  std over  $N=50$  utterances. The F024 demo uses v4; cross-voice generalisation (CSMSC, LJSpeech) is shown with 1 h v2 models and the quality-optimised v4 (5 h,  $\sim 5\times$  data). As UTMOS is wideband-trained, the 16 kHz F024 voice sits low for both resynthesis and reference; the informative quantity is the resynthesis-vs-natural gap within a voice, which is modest throughout and narrows with more data.

## 5. RELATION TO PRIOR WORK

Along the black-box→glass-box spectrum, DDSP [1] learns its filter (timbre, not formant control) and neural

<sup>1</sup>On the unsupervised F024 variant F2→F1 rises to 0.80, an artifact of formant-tracker re-assignment when F2 is driven far below F1 (extreme /y/→/u/), absent with supervised poles or on CSMSC.

Table 1. UTMOS [7] (neural MOS proxy, not a human listening test), mean  $\pm$  std over  $N=50$  utterances: ARIA resynthesis vs. natural reference. “Train” is training-data duration.

| Voice    | SR     | Train | UTMOS resynth   | UTMOS natural   |
|----------|--------|-------|-----------------|-----------------|
| F024     | 16 kHz | 1 h   | $2.53 \pm 0.63$ | $2.70 \pm 0.68$ |
| CSMSC    | 24 kHz | 1 h   | $2.99 \pm 0.39$ | $3.90 \pm 0.37$ |
| CSMSC    | 24 kHz | 5 h   | $3.35 \pm 0.44$ | $3.90 \pm 0.37$ |
| LJSpeech | 22 kHz | 1 h   | $3.43 \pm 0.31$ | $4.37 \pm 0.09$ |
| LJSpeech | 22 kHz | 5 h   | $4.00 \pm 0.23$ | $4.37 \pm 0.09$ |

source-filter models [2] realise the filter with a black-box network. GOLF [4], ARIA’s base, has a differentiable LPC tract but free, learned poles—no analytic formant control. The closest competitor, HiFi-Glot [3], exposes an explicit all-pole tract but with a black-box excitation and a learned parameter→AR mapping. ARIA instead makes both the excitation (a deterministic LF glottal-flow model) and the parameter→filter mapping (analytic biquad placement) deterministic and range-bounded, giving decoupling by construction, a far smaller manipulation error ( $F_2 \approx 20$  Hz vs.  $\approx 150$  Hz reported for HiFi-Glot), and an explicit  $R_d$  glottal voice-quality knob that HiFi-Glot lacks. Relative to GOLF it adds the analytic formant surface, manipulation fidelity as an explicit objective, and a decoupled band-a-periodicity head.

## 6. CONCLUSION AND FUTURE WORK

ARIA is an analytic source-filter neural vocoder that pushes the glass-/black-box boundary to the encoder: the network estimates only interpretable parameters and a deterministic DSP model realises them. The result is control that is decoupled by construction—near-exact for  $F_0$  and  $F_2$ , mutually independent across  $F_0/F_1/F_2$ —learned from  $\sim 1$  h per voice in two languages, and released as an interactive instrument.

The design trades modelling capacity for transparency. Quality still trails the natural reference; spectral tilt and energy co-vary and are not offered as independent axes; the all-pole tract cannot produce anti-formants, so nasals are imperfectly modelled, and  $F_1$  is the weakest knob. The systems are single-speaker. Future work pursues a differentiable pole-zero (ARMA) tract for controllable nasalisation—a qualitatively demonstrable anti-formant knob whose phonetic validation is ongoing—and a low-frequency nasal pole for the  $A_1$ – $P_0$  cue.

## 7. REFERENCES

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